

Lab 2

1. Explain the steps used in the Windows booting process.

Windows Booting Process

The Windows booting process is the sequence of steps that occurs from the moment the computer is powered on until the Windows desktop appears.

1. Power-On and POST (Power-On Self-Test)

When the power button is pressed, the CPU receives power and starts executing instructions stored in the motherboard firmware.

The firmware performs a POST (Power-On Self-Test).

During POST, the system checks hardware components such as:

- RAM
- Keyboard
- Mouse
- Hard Disk/SSD
- Processor
- Display adapter

If any hardware problem is detected, an error message or beep code may be generated.

2. BIOS Initialization

After POST, the BIOS (Basic Input/Output System) takes control.

BIOS initializes hardware devices and determines the boot order.

It searches for a bootable device such as:

- SSD
- HDD
- USB drive
- DVD/CD

Once a bootable device is found, control is passed to the boot loader.

3. Windows Boot Manager Loads

The Windows Boot Manager (BOOTMGR) is loaded from the system partition.

BOOTMGR reads the Boot Configuration Data (BCD) store.

The BCD contains information about:

- Installed operating systems
- Boot options
- Recovery settings

If multiple operating systems are installed, BOOTMGR displays a boot menu for selection.

4. Windows OS Loader Starts

BOOTMGR loads winload.exe, known as the Windows OS Loader.

The OS Loader performs several tasks:

- Loads the Windows kernel (ntoskrnl.exe)
- Loads the Hardware Abstraction Layer (HAL.dll)
- Loads essential system registry information
- Loads boot-start device drivers

5. Windows Kernel Initialization

The Windows Kernel is the core of the operating system.

- It initializes:
- Memory management
- Process management
- Hardware communication
- Security mechanisms

The kernel creates the first system process and prepares the operating system environment.

6. Loading Device Drivers

Windows loads required device drivers for hardware devices such as:

- Keyboard
- Mouse
- Graphics card
- Storage devices
- Network adapters

Drivers allow Windows to communicate with and control hardware components.

7. Session Manager Starts

The Session Manager Subsystem (smss.exe) starts.

It performs important startup tasks:

- Creates system sessions
- Loads system environment variables
- Starts essential Windows subsystems

8. Starting Windows Services

The Service Control Manager (services.exe) starts background services.

Examples include:

- Windows Update
- Network services
- Security services
- Print services

These services help the operating system function correctly.

9. User Logon Process

Windows starts the Winlogon process.

The login screen is displayed.

The user enters credentials such as:

- Username
- Password
- PIN
- Biometric authentication

10. User Profile and Desktop Loading

After successful authentication:

- User profile settings are loaded.
- Startup programs are launched.
- Desktop, taskbar, and icons appear.

The system is now fully operational and ready for use.

Short Flow

Power On → BIOS → Boot Manager → Windows Kernel → Drivers → Services → Login Screen → Desktop

2. Explain the steps used during restart and shut down.

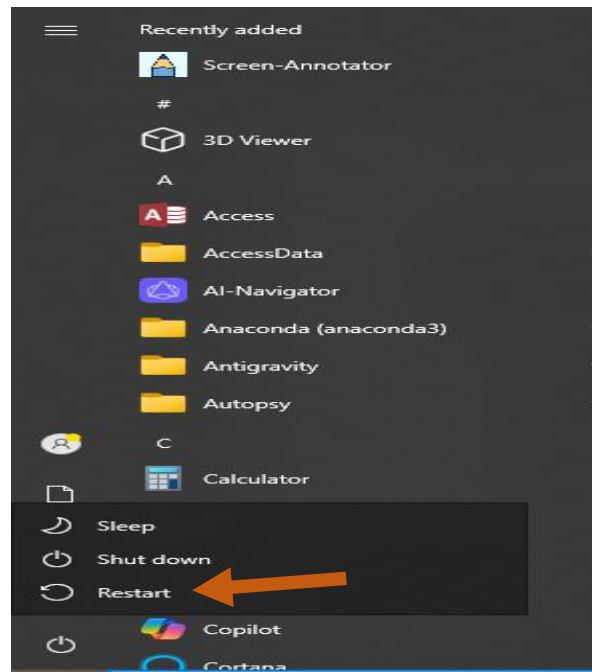
A. Restart Process

A **restart** means the computer turns off and then automatically starts again.

Steps:

1. User Selects Restart
The user clicks Restart from the Start menu.
2. Save Work and Close Programs
Windows asks programs to save data and close properly.
3. Stop Running Services
Background services and processes are stopped safely.
4. Log Off User
The current user session is closed.
5. Shut Down the System
Windows closes system files and releases memory.
6. Computer Reboots
The computer automatically powers on again.

7. Boot Process Starts
BIOS runs, Windows loads, and the login screen appears.

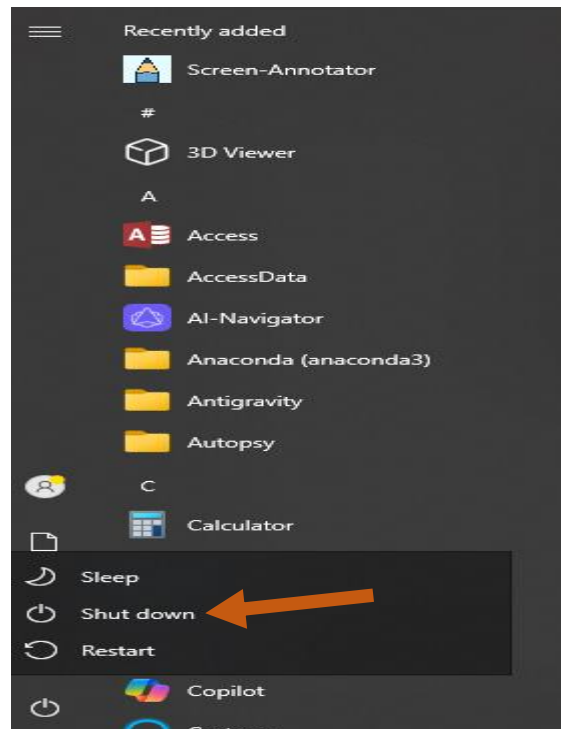


B. Shutdown Process

A shutdown means completely turning off the computer.

Steps:

1. User Selects Shut Down
The user clicks Shut Down from the Start menu.
2. Save Work and Close Applications
Windows requests all running applications to close.
3. Stop Background Services
System services are terminated safely.
4. Log Off User
The user's session is ended.
5. Write Data to Disk
Any remaining data is saved to the hard drive/SSD.
6. Close Operating System
Windows stops all system processes and unloads drivers.
7. Power Off Hardware
The computer receives a command to switch off power.
8. Computer Turns Off
The screen goes blank and the system is completely powered down.



3. Explain the operation of different keys available in a keyboard.

The keyboard contains alphanumeric keys, function keys, control keys, navigation keys, and numeric keys. These keys help users enter data, edit text, navigate documents, and perform computer commands efficiently.

1. Alphanumeric Keys

Used for typing letters, numbers, and symbols.

Includes:

A–Z (alphabet keys)

0–9 (number keys)

Special characters like @, #, \$, %, &, etc.

Example: Pressing A types the letter "A".

2. Function Keys (F1–F12)

Located at the top of the keyboard.

Used for special functions.

Key	Function
F1	Opens Help
F2	Renames a file/folder
F3	Opens Search
F5	Refreshes the screen
F11	Full-screen mode

3. Enter Key

Moves the cursor to the next line.

Confirms commands or selections.

Example: Press Enter after typing a command to execute it.

4. Spacebar

Inserts a blank space between words.

Example: Hello World

5. Backspace Key

Deletes the character to the left of the cursor.

Example: Helloo → Backspace → Hello

6. Delete (Del) Key

Deletes the character to the right of the cursor or removes selected items.

7. Shift Key

Used with another key to type uppercase letters or special symbols.

Example:

Shift + a = A

Shift + 2 = @

8. Caps Lock Key

Makes all letters uppercase until turned off.

Example: HELLO

9. Ctrl (Control) Key

Used with other keys to perform shortcuts.

Examples:

Ctrl + C = Copy

Ctrl + V = Paste

Ctrl + X = Cut

10. Alt Key

Used with other keys for special commands.

Example:

Alt + Tab = Switch between open windows.

11. Esc (Escape) Key

Cancels the current operation or closes menus/dialog boxes.

12. Tab Key

Moves the cursor to the next tab position.

Used to move between fields in forms.

13. Arrow Keys

Move the cursor or selection in different directions:

↑ Up

↓ Down

← Left

→ Right

14. Navigation Keys

Used to move around a document.

Key	Function
Home	Moves to beginning of line
End	Moves to end of line
Page Up	Moves one page up
Page Down	Moves one page down

15. Numeric Keypad

Located on the right side of the keyboard.

Used for quick numeric data entry and calculations.

16. Windows Key

Opens the Start Menu.

Used in shortcuts.

Example:

Windows + E = Open File Explorer

Windows + D = Show Desktop

4. Explain different beep codes of various BIOS.

Beep codes are audio signals generated by the BIOS during POST to indicate hardware problems. Different BIOS manufacturers such as AMI, Award, and Phoenix use different beep patterns to identify faults in components like RAM, CPU, keyboard, video card, and motherboard. These beep codes help technicians diagnose hardware issues when the computer cannot display an error message on the screen.

1. AMI BIOS Beep Codes

Beep Code	Meaning
1 Beep	Memory refresh failure
2 Beeps	Parity error in memory
3 Beeps	Base memory failure
4 Beeps	System timer failure
5 Beeps	Processor (CPU) error
6 Beeps	Keyboard controller error
7 Beeps	CPU exception error
8 Beeps	Display/Video memory error
9 Beeps	BIOS ROM checksum error
10 Beeps	CMOS shutdown register error
11 Beeps	Cache memory error

2. Award BIOS Beep Codes

Beep Code	Meaning
1 Short Beep	System boot successful
2 Short Beeps	POST error

Beep Code	Meaning
1 Long, 1 Short	Motherboard error
1 Long, 2 Short	Video card/display error
1 Long, 3 Short	Keyboard error
Continuous Beeps	RAM or memory problem
Repeated Long Beeps	Memory not installed or faulty

3. Phoenix BIOS Beep Codes

Phoenix BIOS uses a series of beeps separated into groups.

Beep Code	Meaning
1-1-3	CMOS read/write failure
1-1-4	BIOS ROM checksum failure
1-2-1	System timer failure
1-2-2	DMA initialization failure
1-3-1	RAM refresh failure
1-3-3	Base memory failure
1-4-2	Memory error
3-1-1	Master DMA register failure
3-3-4	Video memory failure